

SWYFT Record Twin, status at day 13 of 26

A right-of-way record map goes in. A checked, georeferenced object record comes out, on the LiDAR, with a queue of what could not be proven. This page says where that stands on 2026-09-23, twelve days before the demo.

DATE	2026-09-23
DEMO	2026-10-05, 25 min live
BRANCH	caltrans-spike, loop 5 (2026-09-24 19:15)
DEMO SHEET	R-10434.2, Presidio, SF-101
TEST SET	R-10434.1/.2/.3 (Presidio tile) + R-10741.1/.2/.3 (Marin tile) + gap tile (3DEP 2018) · bench also R-71.71, R-10258.1, R-71.70 · 1969 R-65.1..4
RUN	python spike/demo.py --fast · 128 s · --six 640 s

83% printed distances that match the drawn line 40/48 on R-10434.2 · was 44%	0.04 ft georeferencing RMS, 18 of 19 control points no hand-typed coordinates	44/44 table rows read clean from the CAD text 111/111 keyed cells match	1 figure closed by traverse from the printed record 5 edges · misfit 0.14 ft · area +0.06%
59 items on the exception queue 15 wrong line · 15 no line · the rest genuine or matchline	13 / 19 arc lengths matching the drawn curve was 4/11 · loop 2 · compound curves now split	1 strip tunnel easements 61985-1..4 as drawn one envelope, not four figures · settled, not blocked	6 / 6 sheets georeferenced from their own callouts two tiles · package check ≤ 0.4 ft, offset on none
6 / 6 sheets with LiDAR under them third tile from USGS 3DEP (2018) for the gap · 9 sheets on the bench			

Where we are

Two weeks of spikes proved the pieces. The plan reset on 09-22 named the wall: not reading, but **association**, which drawn line a printed number describes. The 09-23 overnight loop attacked that on the Presidio set with the CAD structure hidden in the PDFs (layers and AutoCAD SHX text annotations) and roughly doubled the pass rates. A second loop the same afternoon made the bench deterministic, tripled the arc-length passes by splitting the long R/W curves where the drawing marks a boundary, and settled the tunnel-easement question: the sheet draws them as one strip, so the four are queued by design, not by failure. What remains before 10-05 is polish and the run of show.

A third loop the same evening (18:00 to 21:10) took the test set from three sheets to six: the Presidio pair plus three sheets on the Marin approach of the Golden Gate Bridge (R-10741.1/.2/.3, 2017, Route 101, drafter CHaldenwang, a Civil 3D 2012-style export with no SHX annotations and no tables, read by OCR). All six are now georeferenced from their own printed callouts and placed on their tiles; Caltrans' packages check all six and are the offset on none. New tile plumbing (`tiles.py`) routes the LiDAR tail per sheet, `demo.py --six` runs the other five sheets through the chain (640 s, 0 failures), and `qgis_project.py --six` builds a two-tile, six-sheet project. Fixing two state bugs made the baseline reproducible: the closing row is `loop3-final`. Coverage: LiDAR lies under four of the six sheets; the bridge and the strait, where the tiles do not meet, need USGS 3DEP. Leg 5 attributed the north misses from 12 crops: the drafter writes the curve delta as `A=` and OCR joins angle and length into one block, so no curve on the north sheets was ever checked. Both are read now: curve $L=R\cdot\Delta$ passes 2 of 2 on R-10741.2 (exact), wrong-line 5 to 3 on .2 and 2 to 1 on .3, no other sheet moved. What is left there has no drawn line to check: chord citations beside curves and a not-to-scale detail inset.

A fourth loop (21:20 to 00:40) went after coverage. A third LiDAR tile from USGS 3DEP (2018, about 10 pts/m²) now lies under R-10741.1 and the east edge of R-10434.3, so every baseline sheet has ground beneath it; R-10434.3 itself turned out to sit 89.5 % on the Presidio tile once its fit was redone, so loop 3's "gap" claim for it was a stale fit. Two more annotation-bearing sheets place from their own control (R-71.71 by grid lines, R-10258.1 by callouts and grid, package agreeing to 0.09 ft), and a third, R-71.70, now takes NGS first-order marks as control: both PIDs fetched, both brass disks found by leader, but the two-point fit disagrees with the sheet's own bearings by 1.29°, so it stays unverified. On the six: the north drafter's dashes were 8.6 to 8.9 pt and the 8 pt cap had been chaining tick marks instead, so the dashed parcel line beside S16°20'26"E now carries its bearing (20/20 on R-10741.2); R-10434.3's line table was masked short by an unrelated curve table at the same x, so 12 table cells had been

checked as labels (wrong-line 45 to 33, every pass kept); comma-grouped distances read. One regression caught by the canonical bench and reverted before it shipped: a sliver guard added for R-71.70 had rerouted 462 real annotations on R-10434.3.

A fifth loop (09-24 evening) went after the south bearings, then the arcs. The bearings had one root cause: a chained line kept its first segment's direction and handed it to every piece cut from it, so every label along a long multi-piece record line was compared with one arbitrary heading. Each piece now takes the direction fitted through its own points, and the three south sheets went from 30/49, 18/31, 32/51 to 34/49, 20/31, 37/52 with nothing lost. The arcs did not move, and the attribution says why: the compound-curve story from loop 2 is down to 8 of 65 misses; 32 are bare distances a length-blind fallback was scoring as arcs (now gated, 7 fake fails gone), and 25 are standalone L= labels with no drawn arc within reach, the next thing to look at.

09-09 to 09-21

Spikes

Text from strokes, OCR, georef from callouts and grid ticks, LiDAR Q1-Q5, scan census. 8 of 11 CAD sheets georeference.

09-22

Plan reset

Association is the wall. Product = the numbers walked as a traverse. Scans = QA story, not input. 13-day roadmap.

09-22

demo.py, objects, QGIS

One command, offline. Object record with provenance. Exception page. Figures and numbers from the run.

09-23 04:00-09:00

Overnight loop

SHX annotation reads, association filters as defaults, traverse edges paired, 1969 scans on the tile. 7 commits, every gate re-measured.

09-23 12:00-14:30

Loop 2

Twin annotations folded. Dash trains on the easement layer. Compound curves split. Arcs 4/11 to 13/19. Easement strip measured as one envelope.

09-23 18:00-21:10

Loop 3

Six zones as the baseline: north tile plumbed, R-10741.1/.2/.3 georeferenced from their own callouts, two-tile QGIS project, bench row loop3-final; north curve data read, wrong-line down.

09-23 21:20-00:40

Loop 4

Coverage: 3DEP gap tile, ground under all six; R-71.71 and R-10258.1 placed; NGS PID control built (R-71.70 unverified, 1.29°); north dashes seen, R-10434.3 table cells masked; bench row loop4-final.

09-24 18:00-19:15

Loop 5

South bearings +11 from one cause (cut pieces inherited the seed direction); arcs re-attributed: fake arc fails gated out, fillets matched by radius; bench row loop5-final.

09-23 now

to 10-05

Day 13

Slides, Q&A sheet, recording not started. Remaining arc misses need a run-sum against a neighbour tag.

Freeze, rehearse

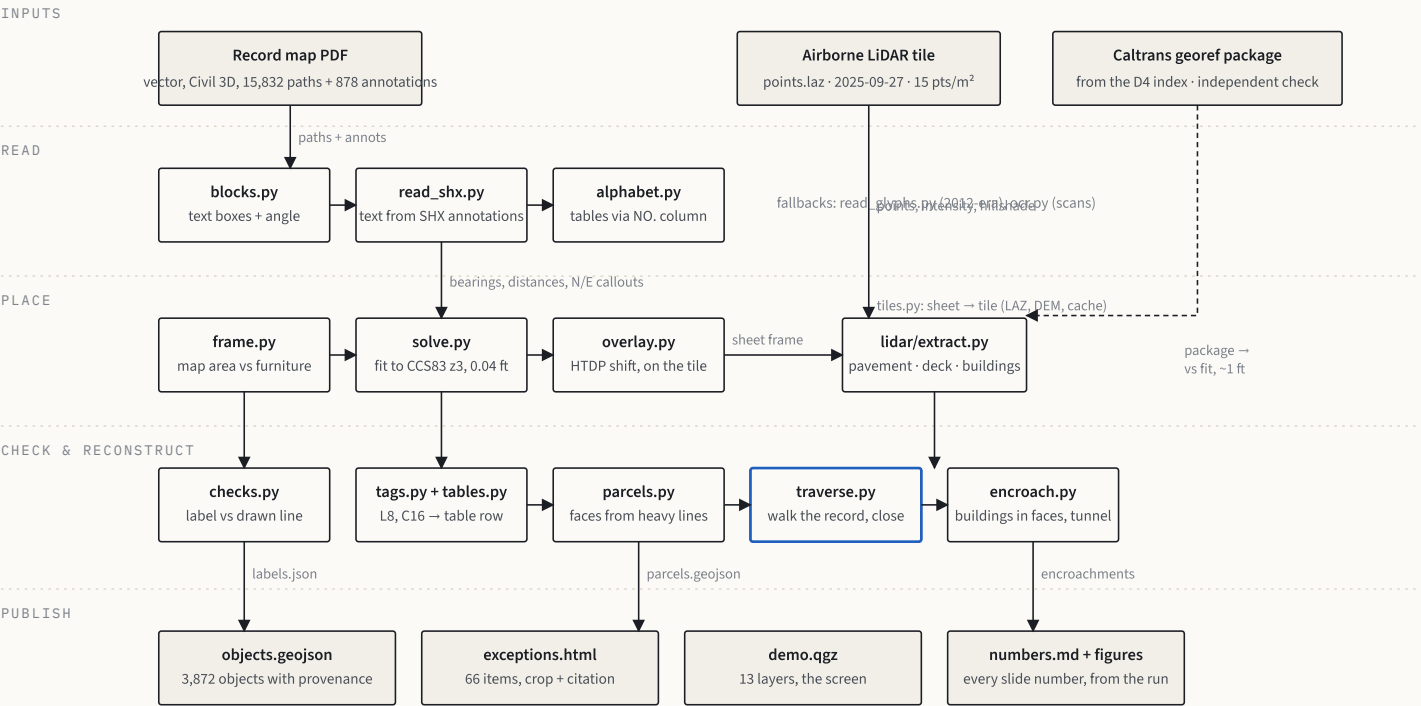
Fallback recording. Two rehearsals. Freeze by 10-03.

Roadmap items

DAYS	DELIVERABLE	STATE	PROOF
1-2	One command, offline, under 3 min. Object record with provenance. QGIS project.	DONE	122 s, 21 steps ok; 3,872 objects each with sheet, rule, residual, status; 13 QGIS layers valid, render PNG as proof
3-6	Table traverse: tags on the drawing tied to table rows, figures walked and closed	DONE AS FAR AS THE SHEET ALLOWS	tags associated 24 of 44 rows, line checks pass; 2 chains close on the demo sheet; the easements are one drawn strip, measured and queued as such (below)
7-8	Exception page a surveyor can work	DONE	one offline HTML, 66 items in cause groups, crop + citation + resolution field per item; scan's queue on the same page
9-10	LiDAR numbers and figures refreshed from the run	DONE	numbers.md and four figures regenerate at the end of every run; tunnel profile caveated until easements close
11	Slides for the non-demo minutes, Q&A sheet, recorded fallback	NOT STARTED	Q&A answers drafted in ROADMAP.md; no recording yet
12-13	Two rehearsals, freeze	NOT STARTED	—
—	Six-sheet baseline (JR, 09-23 evening)	DONE	six.qgz 27 + 21 layers, bench row loop3-final, 5 dispatches

The demo workflow

Everything below runs from `python spike/demo.py --fast` with no network, 129 s. A `--six` flag adds the other five sheets' tails and the six-sheet QGIS project, 640 s, 0 failures. Twenty-one scripts in fixed order, each reading the files the previous ones wrote under `spike/out/`. Read once, then every downstream number is computed from geometry, never re-read.



The pipeline as it runs. Text is read once from the CAD annotations, then every check, closure and area is computed from geometry. The traverse step is the demo's thesis: the printed record walked and closed with no label-to-line heuristic. Caltrans' own package enters only as an independent check on the fit.

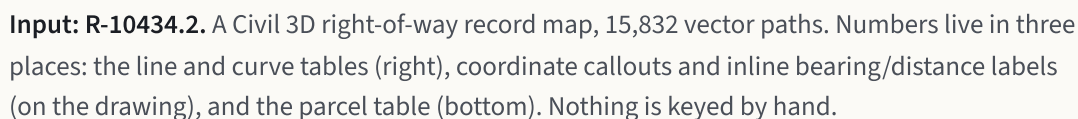
The five demo segments

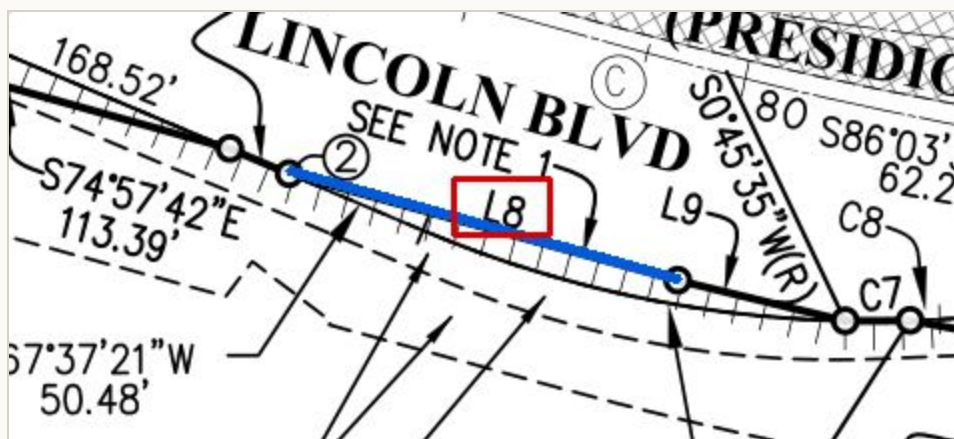
#	SEGMENT	WHAT THE ROOM SEES	STATE
1	Sheet in, objects out	Tables and callouts read; fit residuals; linework on the LiDAR intensity image	LIVE
2	The record checks itself	Tables walked as a traverse; closure per figure; printed vs drawn; exception queue with crops	LIVE
3	Record against ground	Parcels on LiDAR; pavement, deck, building extraction; what lies outside the R/W; HTDP before/after	LIVE

- Two chance scan fits refused on the grid-label scale check;
flightline QA

1969 sheets on the tile via Caltrans' package; 273 boxes, half readable, queue a surveyor would work

Follow segment tag **L8** from the drawing to a verdict. This is the core loop for every table row, every inline label and every traverse edge.

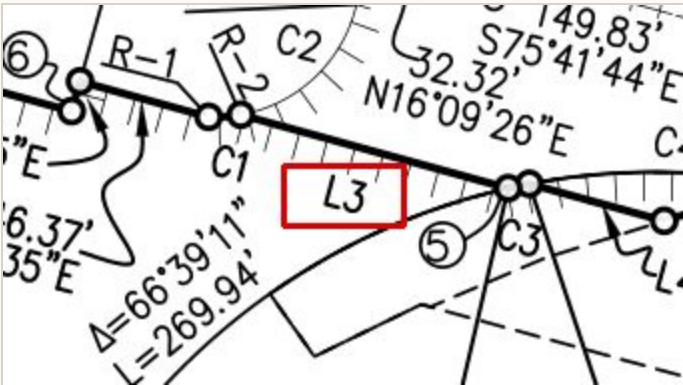




1. Find the tag on the drawing. Tag text from the SHX annotation. The leader's arrowhead picks the segment (blue): a whole piece between vertex circles.

LINE DATA TABLE		
NO.	BEARING	DISTANCE
L1	N18°06'50"E(R)	1.86'
L2	N14°18'16"E	3.76'
L3	S75°11'28"E	96.27'
L4	S75°12'02"E	48.55'
L5	N72°08'15"E	19.34'
L6	S70°11'40"E	6.29'
L7	S74°42'15"E	39.52'
L8	S75°11'28"E	139.90'
L9	S76°56'20"E	59.89'

2. **Read the row.** The NO. column is a known sequence, so the table validates itself. L8 says S75°11'28"E, 139.90 ft.



3. **Or queue it.** L3 has no leader and two candidates. Ambiguity goes to the queue, never a guess.

READ_SHX.JSON · ONE BLOCK PER ANNOTATION

```
{ "cx": 1837.5, "cy": 757.0, "w": 36.0, "h": 7.26,
  "angle": -12.76, "text": "TIE BACK",
  "conf": 1.0, "src": "shx", "id": 2 }
```

TABLES.JSON · L8 AFTER ALPHABET.PY

L8	S75°11'28"E	139.90'
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TAGS_CHECKS.CSV · VERDICT PER ROW

tag	check	printed	drawn	diff	result	assoc
L3	bearing	S75°11'28"E	S75°10'36"E	0.9'	pass	beside
L3	distance	96.27	96.27	-0.00	pass	beside
L4	bearing	S75°12'02"E	S75°12'43"E	0.7'	pass	leader
L4	distance	48.55	48.56	+0.01	pass	leader

GEOREF.JSON · THE FIT EVERY CHECK LIVES IN

```
"crs": "EPSG:2227", "scale_ft_per_pt": 1.38885,
"rotation_deg": 0.536, "rms_ft": 0.041,
"control": [ { "N": 2119812.80, "E": 5993809.52,
               "residual_ft": 0.053, "used": true }, ... ]
```

 Twelve random passing distance labels, each with the label in a red box and the measured line in blue

Output: twelve random passing distance labels, label region in red, the line the reader measured in blue. This is the gate evidence the overnight loop used: a pass only counts if a human sees it on the right line. All twelve are.

Then the record walks itself

TRAVERSE.JSON · ONE CHAIN, CLOSED

```
{ "closed": true, "n_edges": 5, "full_record": 5,
  "misfit_end_ft": 0.14,
  "edges": [
    { "edge": "N67°07'01\"W + 138.21'", "misfit_ft": 0.07},
    { "edge": "L12", "misfit_ft": 0.11},
    { "edge": "L11", "misfit_ft": 0.07},
    { "edge": "N70°37'16\"W + 168.12'", "misfit_ft": 0.10},
    { "edge": "L13", ...}] }
record area 6,970.6 sq ft · drawn face 6,974 · +0.06 %
```



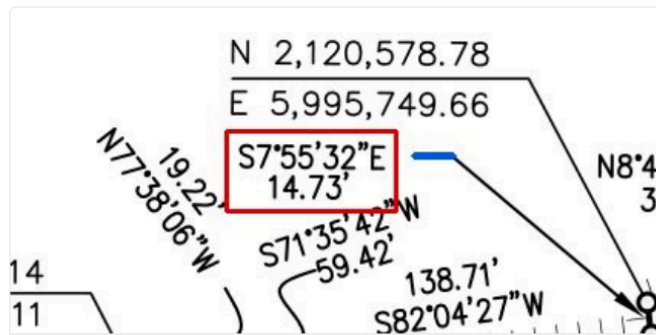
OBJECTS.GEOJSON · ONE OF 3,872 PUBLISHED OBJECTS

```
{ "kind": "control point", "sheet": "r_10434_002_2020-09-16",
  "printed": { "E": 5993809.52, "N": 2119812.80},
  "measured": { "sheet_pt": [797.8, 1026.4] },
  "residual": 0.053, "units": "ft", "status": "verified",
  "rule": "printed N/E callout, leader traced to point,
    similarity fit by consensus; residual vs fit",
  "adjudicated_by": "", "adjudicated_on": "" }
geometry: Point [-122.46557, 37.80025]
```

Every published object carries the sheet, the printed value, the measured value, the residual, the rule that produced it, and an empty adjudication slot. In QGIS a click shows all of it. That is the "reviewer reaches provenance in seconds" promise.

▼ Printed value does not match the line beside it 7

A dimension or bearing disagrees with the line the reader measured (blue) by a small amount. Decide which is the record.



distance

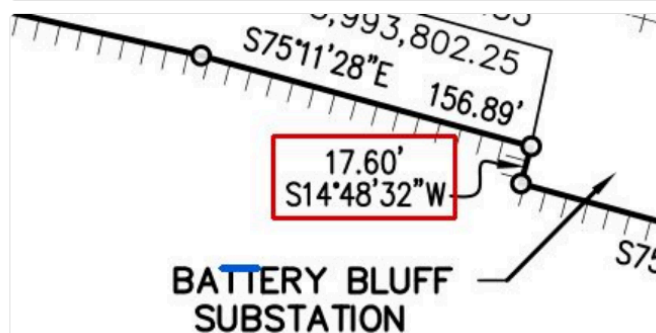
distance: off by +2.60 ft

printed 14.73' · drawn 17.33 ft · +2.60 ft

r_10434_002_2020-09-16 · region 2063, 432, 2118, 458 pt

— resolution —

initials / note



distance

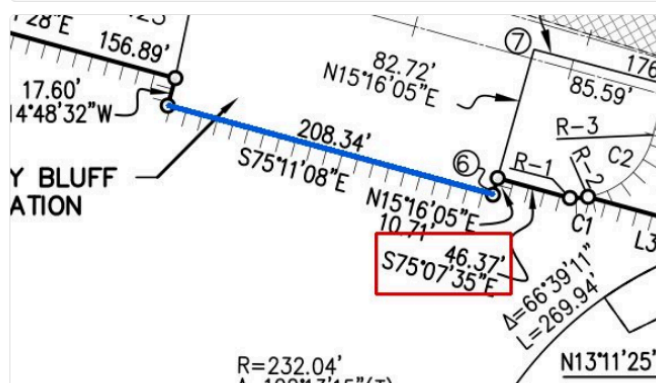
distance: off by -1.35 ft

printed 17.60' · drawn 16.25 ft · -1.35 ft

r_10434_002_2020-09-16 · region 712, 565, 772, 592 pt

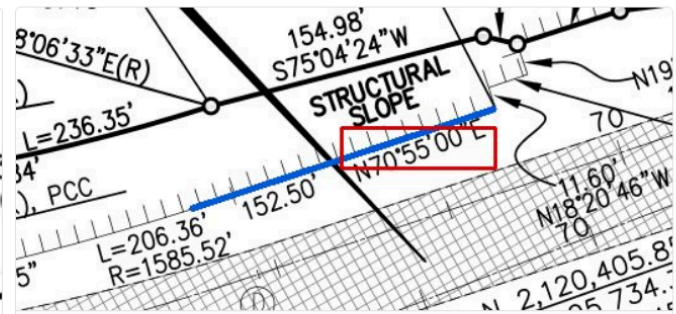
— resolution —

initials / note



bearing

bearing: off by +3.4'



bearing

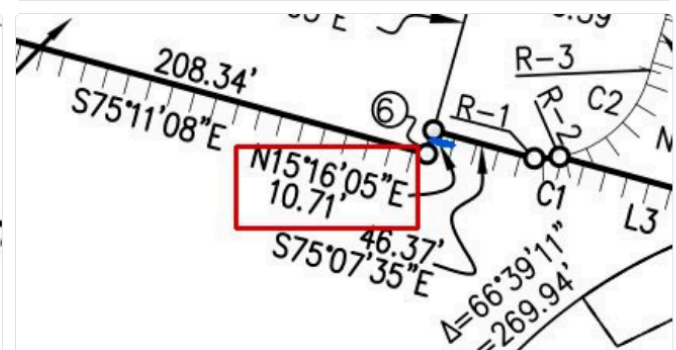
bearing: off by +18.6'

printed N70°55'00"E · drawn N71°13'36"E · +18.6'

r_10434_002_2020-09-16 · region 1992, 558, 2053, 574 pt

— resolution —

initials / note



distance

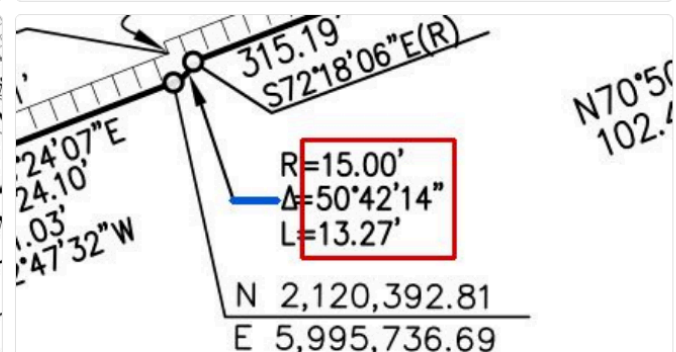
distance: off by -0.75 ft

printed 10.71' · drawn 9.96 ft · -0.75 ft

r_10434_002_2020-09-16 · region 877, 618, 935, 644 pt

— resolution —

initials / note



distance

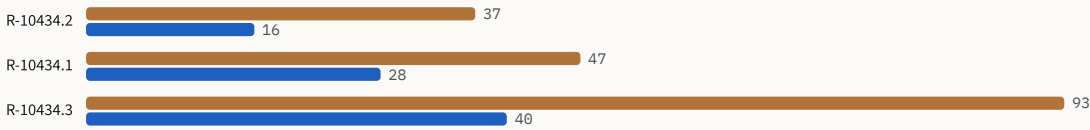
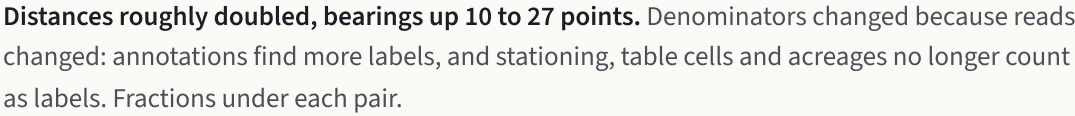
distance: off by +4.58 ft

Output: exceptions.html. One offline file. Each item: crop, cause group, printed vs drawn, citation in PDF points, a resolution a surveyor fills in and exports.

Six sheets, two LiDAR tiles: three Presidio vector sheets by one drafter (R-10434.1/.2/.3, 2020) and three on the Marin approach of the Golden Gate Bridge by another (R-10741.1/.2/.3, 2017, Route 101, no SHX annotations, read by OCR). Every number re-measured by `spike/bench.py loop3-final` on 09-23 at 20:05, not from memory; the row was written twice and matched. R-10434.1 and .3 move from the loop 2 prose (25/32, 41/70 distance passes) to 26/37, 42/70 here because those loop 2 rows were measured on stale sibling state, fixed in loop 3 leg 3; Presidio is unchanged. "Pass" means the printed value agrees with the drawn geometry under the fit: distances to a few hundredths of a foot, bearings to under a minute, on both tiles.

SHEET	TILE	BLOCKS READ	DISTANCE	BEARING	ARC	WRONG - LINE	GEOREF RMS	PACKA
R-10434.2 (Presidio)	South	831/831	40/48	34/49	13/17	14	0.04 ft	~0.1.1
R-10434.1	South	721/721	26/38	20/31	14/30	27	0.02 ft	n
R-10434.3	South + gap	1403/1403	42/59	37/52	23/36	31	0.02 ft	n
R-10741.1	Gap (3DEP)	311/311	14/18	12/13	1/2	3	0.04 ft	0.29 (9 pt ma
R-10741.2	North	381/382	20/22	20/20	0/1	2	0.05 ft	0.37 (9 pt ma
R-10741.3	North	348/349	17/17	13/13	0/1	1	0.03 ft	0.40 (9 pt ma
R-71.71 (2024, SHX)	none on disk	209/209	14/19	20/21	1/5	8	0.01 ft (13 grid lines)	packa on di
R-10258.1 (2020, SHX, metric)	none on disk	434/434	12/49	22/30	0/18	36	0.00 ft (6 callouts + 11 grid lines)	0.09 (9 pt ma

N/M = NOT MEASURED AGAINST THE PACKAGE FOR THAT SHEET. NORTH
"VS PACKAGE" IS THE MAX RESIDUAL AT 9 SHARED SHEET POINTS;
SOUTH IS PRESIDIO'S FIGURE FROM THE SCORECARD BELOW, RASTER-
LIMITED.



Wrong-line, the weakest link, more than halved. On the demo sheet it went 103 (first exception page) to 16. Fixes: leaders by arrowhead, stubs and underlines excluded, runs split at circles and junctions, printed bearing and span as filters.

Full scorecard, R-10434.2 (demo sheet)

MEASURE	RESULT	NOTE
Reading		
Table rows read clean from SHX annotations	44 / 44	R-10434.1 25/25, R-10434.3 83/83. Hand-keyed cells 111/111 match
Text blocks with exact text	832	annotation text, confidence 1.0; no OCR on this sheet class
Georeferencing		

Control points used, RMS	18 / 19 · 0.04 ft	derived scale 1.38885 ft/pt vs plot scale 1.38889
Agreement with Caltrans' own package	~0.6–1.1 ft	raster-limited; the independent check
Printed vs drawn (labels on the drawing)		
Distance	40 / 48	passes agree to 0.03 ft median, 0.24 max
Bearing	34 / 49	passes agree to 0.55' median
Arc length	13 / 17	was 4/11. Long R/W curves now split at circles, meeting lines and radial ticks; (T) totals checked. Misses left: boundaries the drawing does not mark
Curve $L = R \cdot \Delta$ (internal consistency)	10 / 10	
Table rows through their drawing tags		
Rows found on the drawing / associated	33 of 44 / 24	18 by leader arrowhead; 12 partial reads and 33 ambiguities queued, none guessed
Row checks pass · fail	29 · 12	where associated, lines are exact; the arc fails are runs whose sub-arc boundary is not drawn
Reconstruction		
Traverse edges placed / with full record	87 / 49	was 81 / 22 before the loop
Closed chains	2	5 edges, misfit 0.14 ft, area 6,970.6 vs drawn 6,974 sq ft; a second 4-edge loop closes to 0.18 ft
Parcel faces named / within 1 % of the parcel table	8 of 60 / 1	figures leave at matchlines; easement strips do not close
Tunnel easements 61985-1..4	1 strip · 48,410 sq ft	drawn as one envelope; table sum 70,690 sq ft, the TCEs overlap the tunnel easements. QUEUED BY DESIGN
LiDAR		
Features extracted	35 · 11 · 281	pavement, viaduct deck, building polygons (175 pass the roof test)
Highway pavement inside drawn R/W faces	81 %	9,014 m ² outside within 15 m flagged for review

Beyond the demo set

Six-sheet baseline (all six sheets, two tiles)	↑ table above	distance, bearing, arc, wrong-line, georef rms, vs package per sheet — see “Accuracy on the six-sheet baseline”
Vector sheets georeferenced (all D4 vector sheets on disk)	7 / 17	the 10 refused all fail on reads: other drafters' fonts, no tables to validate an alphabet (before the R-10741 set)
Scan R-65.2 (1969, hand-lettered), 32 keyed boxes	16 right · 1 wrong · 15 queued	two readers; not georeferenced by its own text. 48 of 65 sheets in the archive are scans

What the numbers do not claim. The 0.04 ft residual is against numbers read off the same sheet. Independent check is Caltrans' package at about a foot, raster-limited. We claim records-research grade, not a survey tolerance, until a pilot sets one.

The easements, settled

Tunnel easements 61985-1..4 are one drawn strip, not four figures. Loop 2 chained the 345 single-dash strokes on the `rw_EASE_EXIST_align` layer into eight trains. The strip has a north edge drawn under the R/W line, a south edge parallel to it, and closures at both ends. No stroke on the sheet divides it into four. The parcel bubbles' leaders go to R/W vertex circles, and the printed curve data sits on the R/W line itself.

Measured: the envelope is 48,410 sq ft against a table sum of 70,690. The two temporary construction easements overlap the two tunnel easements rather than tiling the strip. Their individual outlines live in the deeds, not on this sheet.

For the demo: segment 2 shows the two chains that close, the strip envelope with its area, and the four easements queued with this reason. That is the story: the system measures what the sheet draws and says what it cannot prove.

Strip region of the sheet with the dashed easement trains coloured, the envelope filled, and the four parcel ovals labelled with their table areas

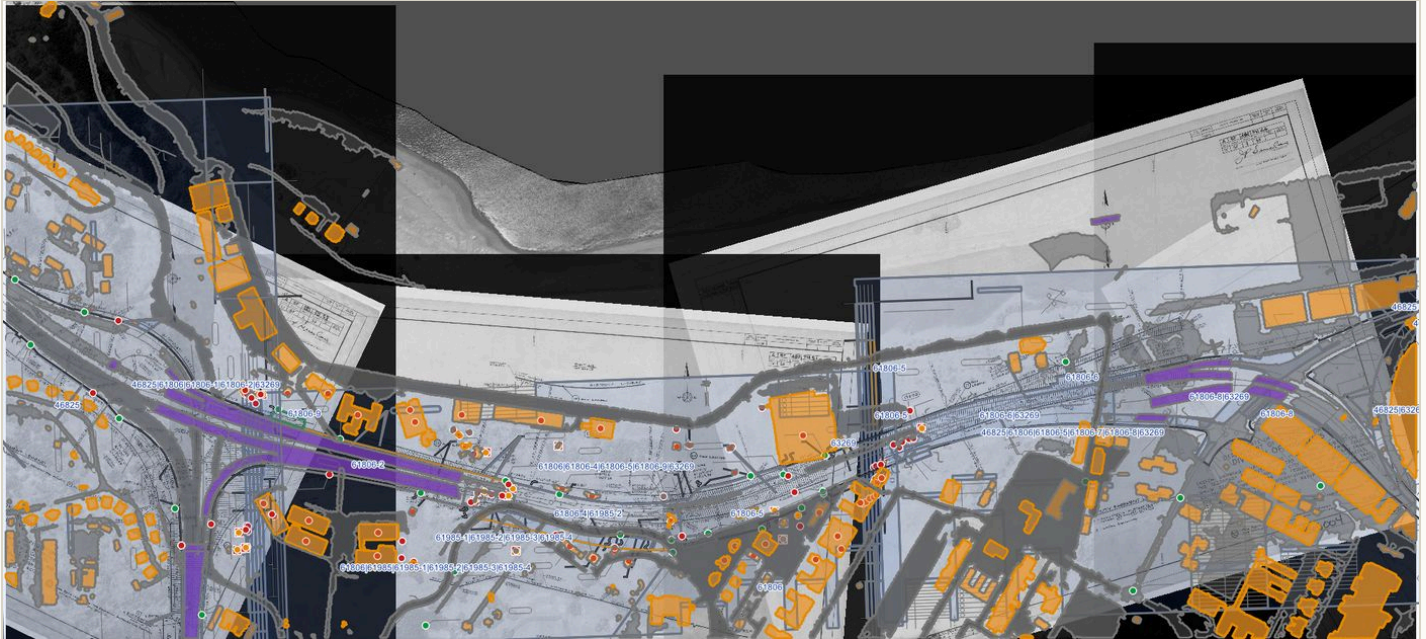
The strip as drawn. Dash trains from the easement layer, the north edge in blue under the R/W line, the envelope filled, the four ovals with their table areas.

The north set

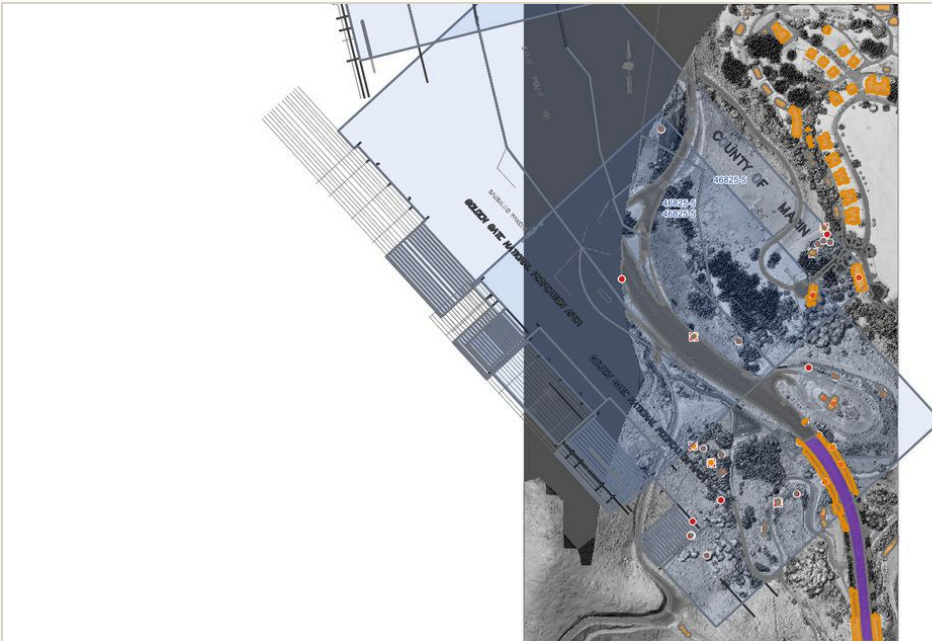
CHaldenwang's callouts read differently from the Presidio drafter's. Coordinates print as a two-row block, N over a bar over E, with a thin (< 1.0 pt) leader running to a filled arrowhead that sits 10.6–10.9 pt short of a small circle rather than touching it. Civil 3D 2012-style export: no SHX text annotations, no tables, so every block is read by OCR. Three callouts paired on R-10741.1, four on .2, three on .3 (10 of 10 traced) once the leader trace learned two rules: walk across thin weights (below 1.0 pt) instead of stopping at them, and snap to a circle within 12 pt of the arrowhead when no second circle sits within 24 pt.

Placement: sheet-derived fits (scale and rotation from the sheet's own bearings and distances, offset from the traced callouts) land at rms 0.04 / 0.05 / 0.03 ft on .1/.2/.3. Checked against Caltrans' package at 9 sheet points: max 0.29 / 0.37 / 0.40 ft. .2 is credible outright; .1 and .3 are weaker fits (3 callouts each) with the rotation confirmed independently by 12 printed bearings apiece. Sheet-derived scale agrees with the package to 0.011 % and rotation to 0.0051° . The package is now the check on all six sheets and the offset on none.

Coverage: the south tile covers R-10434.1 and .2 only — R-10434.3 is 0 % inside it, its nothing starting 5 m past the tile edge (STATE.md's earlier "7 sheets" claim was wrong). The north tile covers R-10741.3 (99.3 %), .2 (59.9 %), .1 (0.3 %). LiDAR lies under four of the six sheets; the bridge and the strait, where neither tile reaches, need USGS 3DEP.



South tile, six.qgz. R-10434.1/.2/.3 linework, parcels and the 1969 record group over the Presidio DEM hillshade and intensity, 27 layers.



North tile, six.qgz. R-10741.1/.2/.3 linework and parcels over the Marin DEM hillshade and intensity, 21 layers.

Next steps

In order. Loop 3 placed all six sheets and dropped leg 5 (north label association) when JR ended the loop, so it leads the list. Items 1 to 5 are code or a package fetch, gated against `bench.py`. Item 6 is the demo itself.

01 Arcs: the 25 standalone L= with no drawn arc in reach

Crop 12 and say whether the arc is drawn at all: a fillet cut into slivers by the split, a curve on a layer the pool skips, a matchline cut. Fix the biggest. Attribution in `spike/out/leg5B_arcs.md`.

02 South bearings, next cause

Leader to another line at a busy vertex (7 of 36): the leader's arrowhead should win over proximity, as it does for table tags.

03 R-71.70's 1.29°

The two NGS marks (AE9850 RYER, AE9865 STAR RESET) fix a frame whose rotation disagrees with the sheet's own bearings by 1.29°; over the 31,700 ft baseline that is ~18 pt on one mark or a non-grid basis of bearings. Crop both marks against the fit and read the basis-of-bearings note. Then NGS control is a second independent check on every sheet that prints a PID.

04 Not-to-scale insets on OCR sheets

The mask is built and text-based, but R-10741.2's caption reads as CJK glyphs, so its two inset labels still fail. Detect the inset by its dashed circle, or fix the OCR crop levelling for small captions.

05 Record residuals on dashed lines

The north dashed parcel line measures 175.04 vs 176.73 printed (1.7 ft), an honest exception; check whether the dashes stop short of the vertex circles by a cap and, if so, extend the train to the mark.

06 Arc lengths, what is left

Compound curves with no drawn mark at the record boundary (the C15/C16 class, L=171.66', C19/C21/C22) need a run-sum against an adjacent tag. Attribution in `spike/out/legC_misses.md`.

07 R-10434.1 traverse

The sibling sheet lost its one closed chain when the arc split moved an edge endpoint. Trace the moved edge.

08 South + gap mosaic; the 2024 San Francisco tile

R-10434.3 should keep the 2025 tile where it has it and take 3DEP only for its east edge. CA_SanFrancisco_B23 (2024) covers both gap boxes but its LAZ carries no CRS; pin it down.

09 The 1950s scans on the north tile

R-71.11/.20/.28 and R-92.8/.9 through their Caltrans packages the same way R-65.1/.3/.4 went onto the Presidio tile.

10 Slides, Q&A sheet, recorded fallback, rehearsals, freeze DAYS 11-13

Non-demo minutes and the scan segment as slides; every number from numbers.md. Record one full run. Two rehearsals inside 25 minutes. No code after the freeze on 10-03.

11 Parked until after 10-05

Scan reading by per-sheet template alphabet (gap 1, the archive's biggest cost). Per-object uncertainty. More drawn-geometry work on 61985-1..4: their outlines are in the deeds.

What has been built

Forty-eight Python scripts, about 8,000 lines, all under `spike/`. Throwaway by design: they answer "does this work" and feed the demo. Grouped by role; pipeline scripts in the order `demo.py` runs them.

FILE	WHAT IT DOES
Pipeline, in run order	
<code>demo.py</code>	Runs the 21 steps in ord offline (<code>--fast</code> , 129 s, and glyph reading wher annotation text covers t Routes QGIS steps to QC Python. <code>--six</code> adds tl five sheets' tails and the QGIS project (640 s, 0 fa
<code>blocks.py</code>	Text blocks from vector strokes: box and angle f piece of lettering, no OC

<code>read_shx.py</code>	Text from AutoCAD SHX annotations, one block annotation. The reader 2016+ sheets. Confidence construction.
<code>frame.py</code>	Where the map is on the map area vs furniture (table block, legends), from the itself.
<code>alphabet.py</code>	A sheet's own alphabet tables: the NO. column in sequence, so its glyphs exemplars. Reads the linear and radial tables.
<code>read_glyphs.py</code>	Fallback reader for sheets without annotations: even matched against the sheet's alphabet. A glyph match
<code>ocr.py</code>	Fallback reader for scan each crop, RapidOCR, page notation.
<code>solve.py + georef.py</code>	One solver over every point control type: N/E callout leaders traced to their point monument notes, grid to Consensus fit to CCS83 : axis redundancy, refuse
<code>tiles.py</code>	Maps a sheet to its LiDAR DEM, cache dir under <code>spike/lidar/cache</code> , the LiDAR tile steps take from it instead of a hard Presidio path.
<code>overlay.py</code>	Linework to NAD83(2011 HTDP epoch shift, draw LiDAR intensity image.
<code>parcels.py</code>	Faces from the heavy line named by parcel labels

	leaders, checked against table areas.
<code>checks.py</code>	Does what the sheet says what it draws? Each bearing distance and arc label a line under the fit; the exception queue.
<code>tags.py + tables.py</code>	Segment tags (L8, C16) and drawing tied to table row leader arrowhead, else a heavy line beside; two columns go to the queue. Per-row
<code>dashes.py</code>	Chains the single-dash segments the easement layer into (dash to dash by heading deduplicated by coverage; checks and parcels seen as lines as geometry.
<code>traverse.py</code>	The record twin: placed edges chained by shared walked by bearing and closure and area where closes.
<code>lidar/extract.py</code>	Pavement, viaduct deck building polygons from returns, deterministical sheet's frame.
<code>encroach.py</code>	Building-class returns in sheet's own R/W and easement faces; terrain profile over tunnel easements.
<code>export_rasters.py</code>	Intensity ortho and hills GeoTIFFs for QGIS.
<code>objects.py</code>	The object record: everything with sheet, printed measured values, residual status, adjudication slot
<code>exceptions_page.py</code>	One offline HTML page: queued item with crop,

printed vs drawn, citatic
resolution field. Scan's
the same page.

qgis_project.py

The screen: PyQGIS buil
project with every layer
ordered, re-reads it, ren
opening view as proof.
builds two tile groups (2
layers) and renders one
tile.

figures.py + slide_numbers.py

One figure per demo seq
rendered from the proje
number the slides quot
run's files.

Measurement and gates

bench.py

The gate instrument: ru
chain on a sheet set and
summarises pass, fail, w
tags, faces, traverse, tak
sheet, plus frame / bloc
bearings_parsed /
distances_parsed over t
sheet set. Every loop leg
accepted or rejected on

gt.py, gt_scan.py

Hand-keyed truth: 227 t
on R-10434.2, self-check
R-Δ; 34 boxes on the 196

sheets.py, batch.py

Generalisation runs: the
chain over every vector
disk and over unrelated
maps from the public in

leg4_crops.py, legB_arcs.py, leg5_61985.py, checks_debug.py

Gate evidence and diag
random passing distanc
and every passing arc re
their lines; the four ease
walked as chains and th
envelope measured; the
line per label drawn.

Scans (most of the archive)

scan_read.py, scan_readers.py, scan_vlm.py	Detect text boxes on a 3 render; RapidOCR and a vision model (qwen2.5v against keyed truth. No values in any prompt.
scan_consensus.py, scan_georef.py	Accept a read only when readers agree or structure it; fit callouts through the solver, refused unless scan matches the grid labels.
scan_on_tile.py	The four 1969 R-65 scan georeferenced by Caltrans packages, HTDP-shifted LiDAR tile as the "1969 r QGIS group.
LiDAR questions answered once	
lidar/q1..q5	Flight date from GPS time (09-27); record on terrain epoch offset by real NGC encroachment; flightline QA (RMSE 0.02–0.04 m c
Documents	
STATE.md, ROADMAP.md, spike/LOOP.md	Resume point with next every known trap; the 1 and the Q&A sheet; the loop's legs, gates and o
spike/out/loop_report.md, legC_misses.md, numbers.md, exceptions.html	The overnight report; the attribution; the slide number queue.

Findings worth carrying forward. Annotation text is the reader for Civil 3D 2016+ sheets; the glyph alphabet stays for 2012-era sheets. CAD layers did not help association once reads were exact; they still separate furniture from map. The exception page's wrong-line group is the fastest diagnostic there is: render the measured line, look at twelve crops, fix the biggest cause, repeat. Two generic readers agreeing on hand lettering is a floor, not a method.

```
SOURCES: STATE.MD, ROADMAP.MD, SPIKE/LOOP.MD,  
SPIKE/OUT/LOOP_REPORT.MD, SPIKE/OUT/LEGC_MISSES.MD,  
SPIKE/OUT/NUMBERS.MD, SPIKE/OUT/BENCH.CSV (ROWS "LOOP5-  
FINAL") · EVERY NUMBER RE-MEASURABLE WITH PYTHON  
SPIKE/BENCH.PY FINAL --TABLES --TAGS --PARCELS --TRAVERSE  
PRESIDIO R10434_1 R10434_3 R10741_1 R10741_2 R10741_3 R71_71  
R10258 R71_70
```